

Article

Monocular Estimation of Grape Berry Size (Caliber) Distributions Using Geometry-Aware Representations and Structured Prediction

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Abstract

Grape caliber distributions are critical for packing, grading, yield estimation, and post-harvest logistics. However, estimating reliable caliber histograms from single images remains challenging due to occlusion and dense bunch structure. This work presents a two-stage monocular pipeline that integrates instance segmentation, geometry-aware representations, residual quantity correction, and structured histogram prediction. In the first stage, a YOLO-based model detects grape instances and a calibration object, enabling the construction of geometry-aware auxiliary channels and a segmentation-derived counting prior. In the second stage, these representations are used to estimate total grape count and caliber distributions. Results show that RGBDT consistently outperforms RGB, indicating that geometry-aware cues improve both histogram fidelity and counting accuracy. The framework achieves stable performance under realistic conditions while maintaining low runtime, supporting practical deployment in agricultural environments.

Keywords: grape caliber estimation; grape caliber histogram prediction; instance segmentation; monocular vision; distance transform; residual count estimation; lightweight deep learning; ONNX deployment

1. Introduction

1.1. Problem Context and Motivation

Grape caliber, defined as the diameter of individual grape berries, plays a critical role in viticulture and post-harvest operations, as berry-size distributions directly influence packing, grading, yield estimation, and logistics planning [1]. However, in real-world environments, grape bunches are captured under variable illumination, cluttered backgrounds, viewpoint changes, and self-occlusion, which significantly complicates accurate estimation from a single image. Consequently, relying solely on scalar descriptors such as mean berry size is insufficient for many agricultural applications. Instead, distribution-level representations, particularly caliber histograms, provide richer information by capturing variability, skewness, and concentration across discrete size classes [2]. Therefore, this study focuses on predicting the complete caliber distribution, reflecting a deliberate effort to align computational modeling with practical decision-making requirements in viticultural workflows.



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1.2. Limitations of Existing Approaches

Existing vision-based solutions for grape measurement typically rely on controlled acquisition setups, depth sensors, stereo rigs, or multi-view systems, which can deliver accurate metric estimations [3]. However, these approaches introduce significant practical constraints, including increased hardware cost, complex calibration procedures, synchronization requirements, and reduced portability. As a result, their applicability in real-world agricultural environments remains limited [4]. In contrast, field conditions demand solutions that are robust to variability, simple to deploy, and computationally efficient. The reliance on specialized acquisition systems restricts scalability and hinders widespread adoption, motivating the exploration of lightweight, monocular alternatives better aligned with operational constraints in viticulture.

1.3. Challenges of Monocular Estimation

Monocular RGB imagery provides a practical and scalable alternative for grape analysis; however, it introduces significant challenges related to scene complexity and limited visual information [5]. In particular, dense bunch structures produce frequent self-occlusion, which hinders reliable counting and diameter estimation. Moreover, a single viewpoint offers only partial evidence of the true caliber distribution, while neighboring berries in clustered regions often appear visually ambiguous. Thus, these factors complicate the inference of both total grape count and the full histogram of calibers from RGB appearance alone. Therefore, addressing these limitations requires approaches that can compensate for missing information and ambiguity inherent in monocular observations [6].

1.4. Opportunity: Learning-Based Monocular Pipeline

Recent advances in lightweight instance segmentation and deployable deep learning models offer new opportunities to address monocular grape analysis more effectively [7]. Reliable detection of grape instances and a calibration object can support segmentation outputs to serve not only for localization but also as structured intermediate representations. Consequently, these representations may encode spatial relationships and approximate geometric cues that are otherwise implicit in RGB data [8]. Therefore, a two-stage design is motivated, where segmentation-derived features provide geometry-aware inputs and an initial count prior, which are subsequently refined by dedicated models for total-count correction and histogram estimation. This approach reflects a deliberate effort to decouple perception from structured prediction while maintaining efficiency and deployability.

1.5. Proposed Approach

This work proposes a two-stage monocular pipeline for grape caliber analysis from single images, combining instance segmentation with structured prediction to address limitations of RGB-only inference [5]. In the first stage, a YOLO-based model detects grape instances and a calibration object, enabling the construction of geometry-informed feature channels via distance transforms and a segmentation-derived counting prior [9]. Subsequently, these representations are processed by a residual quantity model and a histogram model to estimate total count and normalized caliber distribution, which are combined to reconstruct the final histogram [10]. Furthermore, RGB and RGBDT modalities are compared across multiple histogram-modeling families to assess representational and parametric effects. Importantly, the two-stage design reflects a deliberate effort to decouple geometric reasoning from distribution inference, improving robustness, interpretability, and alignment with practical workflows requiring full distribution estimation.

1.6. Hypotheses

This study investigates whether monocular grape caliber estimation can be improved by incorporating geometry-aware representations derived from segmentation outputs rather than relying solely on RGB appearance. In particular, the proposed framework evaluates the role of structured intermediate representations and modeling choices in enhancing both distributional and counting accuracy.

Hypothesis 1. *Augmenting RGB input with segmentation-derived geometric cues, specifically grape and ping-pong distance-transform channels, improves both caliber histogram prediction and total-count estimation compared to RGB-only representations.*

Hypothesis 2. *Leveraging a segmentation-derived counting prior with residual correction, combined with flexible histogram formulations such as direct-bin prediction, provides more accurate and robust estimation of grape count and caliber distributions than direct regression or strongly constrained parametric models.*

For completeness, the corresponding null hypotheses are defined as follows: (i) the inclusion of geometry-aware channels does not produce measurable differences in histogram or count estimation compared to RGB-only input, (ii) residual correction does not improve total-count estimation relative to direct prediction approaches, and (iii) different histogram-modeling strategies do not produce statistically distinguishable performance under the evaluated protocol.

1.7. Research Questions

The proposed framework introduces a geometry-informed, two-stage framework for monocular grape caliber estimation, combining segmentation-derived representations with structured prediction models. To systematically evaluate its effectiveness and practical relevance, the study is guided by a set of research questions that address performance, representation design, and modeling strategies. In particular, the following questions aim to assess the capability of the pipeline under realistic conditions, the contribution of geometry inputs and residual learning, and the suitability of different histogram-modeling approaches for capturing complex caliber distributions while maintaining deployment feasibility.

RQ1: To what extent can a two-stage monocular pipeline estimate grape caliber histograms and total grape count from single images under realistic acquisition conditions?

RQ2: Does the incorporation of segmentation-derived geometric representations, together with residual correction of a counting prior, improve performance compared to RGB-only input and direct estimation approaches?

RQ3: Which histogram-modeling strategy provides the best trade-off between flexibility, distributional fidelity, count consistency, and deployment efficiency in practical settings?

1.8. Novelty

This work addresses the limited availability of practical monocular methods for estimating grape caliber distributions from single images under lightweight deployment constraints. Specifically, the proposed approach integrates instance segmentation, geometry-aware channels derived from distance transforms [11], residual quantity correction, and histogram prediction within a unified two-stage framework. Rather than relying solely on RGB appearance, the method leverages segmentation-derived representations and a counting prior to enhance both distributional fidelity and total-count estimation. Furthermore, systematic evaluation across RGB and RGBDT inputs and multiple histogram-modeling families may improve a structured analysis of how representation and parameterization affect performance and deployability. From a conceptual standpoint, the formulation re-

frames monocular perception as structured inference under partial observability, where intermediate geometric representations act as latent variables that regularize the mapping between visual input and distributional outputs, improving robustness and generalization across dense object arrangements.

1.9. Contributions

The contributions of this work are fourfold. First, a two-stage monocular pipeline is proposed, integrating instance segmentation, geometry-informed feature construction, residual quantity estimation, and histogram prediction for grape caliber analysis. In this context, “caliber” denotes berry diameter expressed in millimeters and used to define size classes in grading and distribution analysis. Second, an RGBDT representation is introduced and evaluated, extending RGB input with grape-instance and ping-pong distance-transform channels derived from segmentation outputs to enhance geometric awareness. Third, multiple histogram-modeling families are systematically compared under a unified experimental setting, enabling a structured evaluation of both constrained and flexible output parameterizations for caliber distribution prediction. Finally, the proposed framework is analyzed from both accuracy and deployment perspectives, including total-count estimation, ONNX export fidelity, and end-to-end runtime, reflecting a deliberate effort to support practical, lightweight, and scalable grape bunch analysis in real-world conditions.

1.10. Paper Organization

The remainder of this paper is organized as follows. Section 2 reviews prior work on vision-based fruit sizing, instance segmentation for agricultural measurement, reference-based scale recovery, and structured histogram prediction. Section 3 describes the proposed two-stage framework, including the segmentation stage, the construction of geometry-aware representations, and the histogram and total-count prediction models. Section 4 presents the experimental evaluation, including segmentation performance, comparison across input modalities and histogram-modeling families, and deployment-oriented analyses. Section 5 discusses the main findings, limitations, and practical implications of the proposed approach. Finally, Section 6 summarizes the contributions and outlines directions for future work.

2. Related Work

This section reviews prior work related to vision-based fruit sizing, instance segmentation for agricultural measurement, reference-based scale recovery, and structured output prediction. The discussion highlights the limitations of existing approaches under real-world conditions and motivates the need for geometry-aware, monocular pipelines capable of reliable distribution-level inference and efficient deployment.

2.1. Vision-Based Fruit and Grape Sizing

Classical approaches to fruit sizing rely on geometric measurements derived from controlled RGB imaging, often assuming uniform backgrounds, fixed camera distances, and minimal occlusion. More recent methods incorporate depth sensors or stereo vision to recover metric scale and improve diameter estimation accuracy [12]. Learning-based approaches using convolutional neural networks have further improved detection and segmentation performance; however, many still depend on structured acquisition setups or multiple viewpoints [13]. In the context of grape analysis, dense clustering and frequent occlusions pose additional challenges, making it difficult to generalize these methods to field conditions where lighting, viewpoint, and scene complexity vary significantly.

2.2. Instance Segmentation for Agricultural Measurement

Instance segmentation has emerged as a powerful tool for agricultural measurement tasks, enabling precise localization and delineation of individual objects such as fruits, leaves, and clusters. Architectures such as Mask R-CNN and YOLO-based segmentation models have been successfully applied to crop monitoring, yield estimation, and phenotyping [14]. These models provide pixel-level masks that can be used to derive geometric features, including size and shape descriptors. However, segmentation accuracy may degrade under occlusion, overlapping instances, and complex backgrounds, which are common in grape bunches [15]. Moreover, while segmentation facilitates object detection, it does not directly solve the problem of inferring complete distributions from partial observations.

2.3. Physical References for Metric Scale Recovery

Recovering real-world scale from monocular images is a long-standing challenge in computer vision. One practical solution involves the use of physical reference objects with known dimensions, such as calibration markers or standardized objects placed within the scene [16]. These references allow the estimation of scale factors and geometric transformations that map image measurements to real-world units. In agricultural applications, reference-based methods have been used for fruit sizing and plant measurement under field conditions [17]. Nevertheless, their effectiveness depends on consistent visibility, accurate detection, and proper geometric modeling, and they often do not address distortions caused by perspective or non-planar arrangements.

2.4. Histogram and Structured Output Prediction

Predicting structured outputs such as histograms has gained relevance in tasks where aggregate properties are more informative than individual predictions [18]. In particular, methods based on direct bin regression, parametric distributions, and constraint-aware learning have demonstrated effectiveness in controlled environments. However, under partial observability, including occlusion and incomplete views, models must infer missing information while preserving global consistency [19]. Consequently, techniques such as distribution matching and auxiliary supervision have been explored, yet their application to dense agricultural scenes remains limited due to variability and scarce annotations. Despite these advances, existing approaches either emphasize geometric estimation or detection and counting, rarely addressing full histogram prediction under realistic monocular conditions [20]. Therefore, this gap motivates the development of a modular, geometry-aware pipeline that bridges perception and distributional inference.

2.5. Limitations for Edge Deployment

Despite significant advances in vision-based measurement and deep learning, many existing solutions are not optimized for edge deployment. High computational requirements, large model sizes, and reliance on specialized hardware limit their applicability in real-time and resource-constrained environments [21]. Additionally, robustness to environmental variability, including lighting changes and occlusions, is often insufficient for field deployment. Lightweight architectures and model compression techniques, such as quantization and pruning, have been proposed to address these issues [22], yet achieving a balance between accuracy, efficiency, and scalability remains an open challenge. These limitations motivate the development of approaches specifically designed for low-power, real-time agricultural applications.

2.6. Research Gap and Contribution

Despite prior advances, a fundamental gap remains at the intersection of monocular perception, structured distribution prediction, and deployable system design. Specifically, existing approaches often address geometric estimation, detection and counting, or efficiency in isolation, while the joint problem of inferring complete caliber distributions under realistic conditions remains insufficiently explored [23]. To address this, a modular two-stage framework is introduced, integrating instance segmentation, geometry-aware representation learning, and structured histogram prediction. Rather than focusing on object-level outputs, the approach leverages segmentation-derived representations as intermediate variables to support distribution-level inference [24]. This enables estimation of full-caliber histograms while jointly modeling count scale and distribution shape through residual correction. Furthermore, systematic evaluation across input modalities and modeling strategies, combined with deployment-oriented validation, positions the contribution as a coherent and practical framework for monocular grape caliber analysis.

3. Materials and Methods

This section presents a framework for grape caliber histogram prediction and total-count estimation from monocular imagery. Specifically, the approach is organized into four components: the system pipeline for model construction, the inference pipeline defining operational behavior, the modeling formulation, and the design rationale guiding architectural choices. In addition, the dataset consists of monocular RGB images acquired under realistic vineyard conditions, capturing variability in illumination, viewpoint, and occlusion. Each sample includes a grape bunch and a ping-pong reference object for calibration. The dataset covers multiple grape varieties and colors, with several hundred annotated images and thousands of grape instances, exhibiting class imbalance toward individual grapes. All samples were manually annotated using an instance segmentation protocol, enabling both geometry-aware representation learning and structured histogram prediction under practical conditions.

3.1. System Pipeline

The system pipeline describes the process used to construct, train, and deploy the models involved in the proposed framework. This includes dataset annotation, segmentation model training, generation of intermediate representations, and subsequent training of prediction models. The overall system pipeline, including data preparation, model training, and deployment stages, is illustrated in Figure 1.

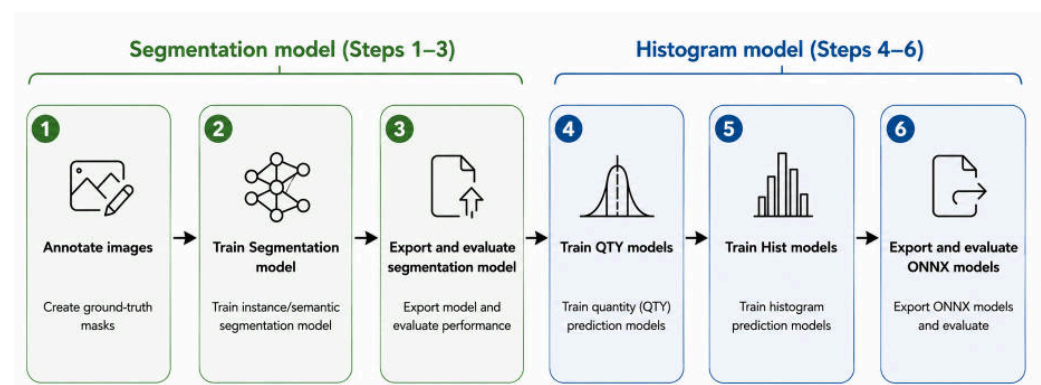


Figure 1. Training and deployment workflow of the proposed methodology. The pipeline is divided into two stages: segmentation and histogram prediction. In the first stage, annotated images are used to train a segmentation model, which is then exported and applied at runtime to generate cached masks

and derived representations. In the second stage, these representations are used to train residual quantity and histogram prediction models, which are subsequently exported and evaluated in ONNX format for efficient deployment.

This workflow separates perception from structured prediction by generating fixed segmentation-derived inputs that are reused across experiments. Such decoupling ensures consistent feature extraction, improves reproducibility, and enables controlled comparison between histogram-modeling strategies.

3.2. Inference Pipeline

The inference pipeline defines how the proposed method operates on a single input image at test time. The overall process is illustrated in Figure 2.

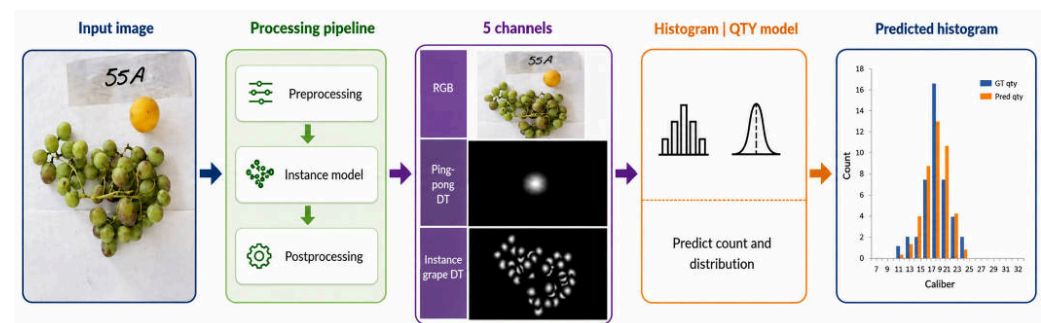


Figure 2. Overview of the proposed monocular pipeline for grape caliber histogram prediction. A single RGB image is processed through three sequential stages: (1) perception, where preprocessing, instance segmentation, and postprocessing extract grape instances and a reference object; (2) geometric encoding, where segmentation outputs are transformed into geometry-aware representations, including RGB and distance-transform channels; and (3) prediction, where a histogram model and a residual quantity module jointly estimate the caliber distribution and total grape count. The design emphasizes a single-input, feature-transformation pipeline suitable for robust estimation under partial visibility and real-world conditions.

Given a single image of a grape bunch, the objective is to estimate both the total number of berries and the full-caliber histogram over the range from 7 to 32 mm. First, a segmentation model detects grape instances and the ping-pong calibration object. Second, the segmentation output is converted into geometry-aware auxiliary maps, specifically a grape-instance distance transform and a ping-pong distance transform. Third, these representations are used as input to two prediction branches operating in parallel: a histogram prediction model that estimates the normalized caliber distribution, and a residual quantity model that refines the total count by correcting the segmentation-derived prior. The final predicted count per caliber bin is obtained by combining both outputs.

3.3. Modeling Formulation

From a modeling perspective, the proposed method can be interpreted as a structured decomposition of the inverse problem associated with monocular perception. The first stage approximates a latent geometric representation of the scene through instance segmentation and distance-transform encoding, effectively lifting the input from raw RGB appearance to a spatially structured domain. The second stage operates on this representation to infer global properties, namely the caliber distribution and total count, which are not directly observable from the input.

This formulation enables the use of intermediate supervision and reduces the complexity of the mapping between input and output spaces. By operating on structured representations rather than raw images alone, the model benefits from improved stability

and enhanced interpretability, particularly in scenarios with occlusion and dense object arrangements. The comparative evaluation focuses on variations of the proposed framework under controlled conditions. While the design enables consistent comparison across input modalities and histogram-modeling strategies, it does not include a direct baseline performing count estimation solely from raw image features. Therefore, the reported comparisons should be interpreted within the scope of the defined experimental framework.

3.4. Design Rationale

The adoption of a two-stage formulation is motivated by the inherent difficulty of jointly learning geometric structure and distributional outputs from raw image data. In dense and occluded scenes, such as grape bunches, end-to-end approaches must implicitly learn both instance separation and global aggregation within a single mapping, increasing model complexity and sensitivity to noise.

By decoupling perception from structured prediction, the proposed design allows segmentation to specialize in spatial delineation while the prediction stage focuses on distribution-level inference. This separation improves learning efficiency, facilitates error analysis, and enables modular experimentation. Furthermore, it supports practical deployment by allowing lightweight segmentation and prediction models to be independently optimized and exported for edge inference.

3.5. Stage 1: Instance Segmentation

3.5.1. Instance Segmentation Dataset

An instance segmentation dataset was curated to support reliable grape localization and physical reference detection under realistic acquisition conditions. Each image contains one grape bunch and one ping-pong ball acting as a metric reference. The annotation scheme follows a multi-class instance format including pingpong, bunch_black, bunch_green, bunch_red, grape_black, grape_green, and grape_red. The annotation distribution is strongly imbalanced, with grape instances dominating the label mass and comparatively fewer bunch-level annotations. The class distribution of the annotated dataset is summarized in Figure 3.

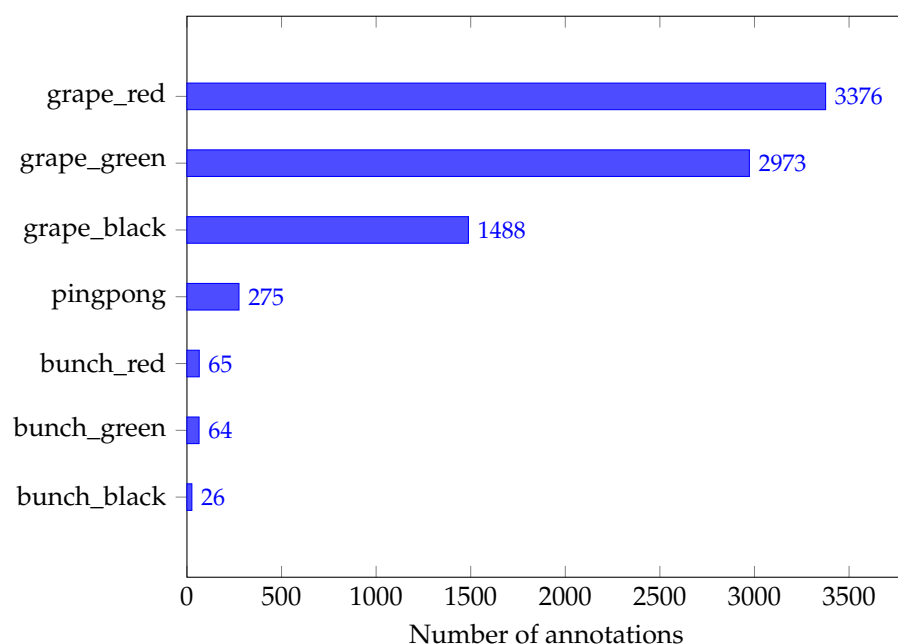


Figure 3. Distribution of instance annotations across classes in the segmentation dataset. Grape-level categories dominate the dataset, reflecting the dense object structure of grape bunches, while bunch-

level annotations are comparatively scarce. This imbalance highlights the need for robust instance-level detection under highly skewed label distributions and motivates the use of geometry-aware representations to support downstream tasks such as counting and histogram estimation.

3.5.2. Instance Segmentation Model and Training

Instance segmentation was performed using a YOLO-based segmentation model operating at 512×512 input resolution. The model predicts masks for grape instances, bunch regions, and the ping-pong reference. These masks constitute the first stage of the proposed pipeline and provide the geometric information required by the downstream histogram and quantity models.

The segmentation model was trained on the curated multi-class instance dataset described above. During optimization, training progress was monitored through the evolution of the segmentation losses and the corresponding box- and mask-based metrics across epochs. After training, the segmentation model was exported for deployment, preserving the mask-level predictions required by the complete inference pipeline.

3.5.3. Segmentation-Derived Inputs and Base Counting Prior

Before training the histogram and quantity models, each image is first processed by the segmentation model. The predicted masks for `grape_green`, `grape_red`, and `grape_black` are used to build the grape distance-transform channel, whereas the predicted pingpong mask is used to generate the calibration-object distance-transform channel. In parallel, the number of detected grape instances is computed and stored as the segmentation-derived base count, denoted by `seg_count_base`.

To avoid recomputing segmentation at every training epoch, the segmentation outputs and derived representations are cached in .npz format. Each cache stores the resized RGB image, the grape distance transform, the ping-pong distance transform, and metadata such as the number of grape detections, the number of ping-pong detections, the original image size, and letterbox information.

3.6. Representation and Dataset Construction

3.6.1. Histogram Dataset and Ground-Truth Targets

The histogram dataset is organized as a structured table in which each row corresponds to a single-image capture containing one bunch and one ping-pong reference. The schema stores image identifiers; grape attributes such as variety, color, form, and size; total grape count; minimum and maximum caliber; and per-bin caliber counts. For the histogram prediction experiments, adjacent half-millimeter caliber annotations are aggregated into integer-millimeter bins, producing the discrete target histogram used for supervision. In parallel, the histogram-derived mean caliber is used as an additional supervisory signal for the histogram models, while the total grape count is used to supervise the residual quantity model. A representative example of the dataset, including the image capture and associated annotation context, is shown in Figure 4.

It is important to note that, although the dataset captures variability in grape variety, color, and bunch structure, it may still reflect biases associated with specific acquisition conditions and sampling strategies. In particular, the distribution of calibers and total counts is not uniform, with a concentration of samples in intermediate ranges, as observed in the exploratory analysis. This imbalance may influence model training by favoring more frequently observed configurations. Therefore, the reported results should be interpreted within the context of the dataset characteristics, and future work should consider broader data collection to improve generalization across diverse vineyard conditions.

3.6.2. Dataset Partitioning and Experimental Protocol

The dataset used in the deep learning experiments was partitioned at the bunch level using `grape_id` as the grouping key, ensuring that images from the same bunch were never shared across training and validation. The resulting experimental setup comprised 7503 training samples and 1883 validation samples. Variety names were normalized, yielding 12 distinct varieties encoded as categorical indices. For model comparison, a fixed validation subset was additionally constructed by selecting one sampled view per bunch, resulting in 944 validation examples.

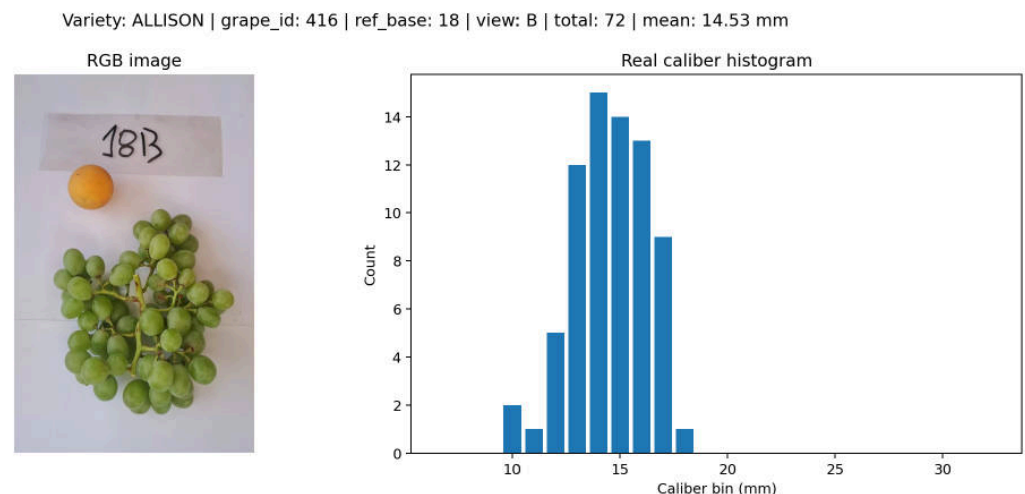


Figure 4. Example of a dataset sample used for histogram prediction. Each capture contains a single grape bunch and a ping-pong ball serving as a physical reference for scale recovery. The associated annotations enable the derivation of per-bin caliber distributions, total grape count, and auxiliary attributes, forming the basis for supervised learning of both histogram prediction and quantity estimation under realistic acquisition conditions.

3.6.3. Dataset Exploration

An exploratory data analysis (EDA) was conducted to characterize the main statistical properties of the dataset before model training. This analysis considered the partition sizes, the number of unique bunches per split, the variety distribution, the aggregate caliber distribution, the total grape count distribution, the mean caliber distribution and the relationship between total count and mean caliber.

Figure 5 shows that the dataset is imbalanced across grape varieties, that caliber mass is concentrated in intermediate bins, and that total grape count exhibits broad variability. These observations provide important context for the prediction task.

3.6.4. Input Representation

Two input modalities were considered to evaluate the impact of geometry-aware representations on model performance.

RGB

The `rgb` modality uses only the three original color channels, providing a baseline representation based solely on appearance information.

RGBDT

The `rgbdt` modality extends the RGB input by incorporating two additional geometry-aware channels derived from segmentation outputs: the grape-instance distance transform

and the ping-pong distance transform. These channels encode spatial structure and proximity information, complementing the visual features present in the RGB image.

Accordingly, the model input is defined as

$$\mathbf{x} \in \mathbb{R}^{3 \times H \times W}$$

for the RGB modality, and

$$\mathbf{x} \in \mathbb{R}^{5 \times H \times W}$$

for the RGBDT modality, with channel ordering given by

$$\{R, G, B, DT_{grape}, DT_{pp}\}.$$

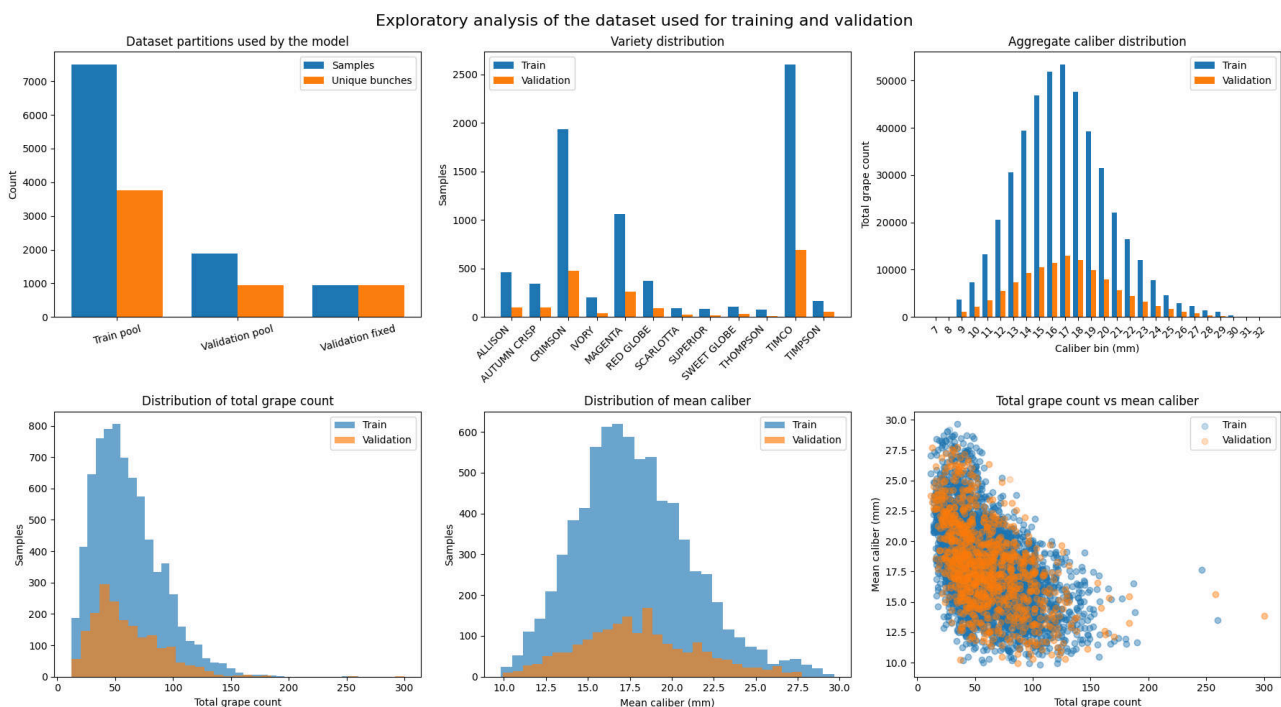


Figure 5. Exploratory analysis of the curated dataset used in this study. The figure summarizes the train and validation partitions, variety distribution, aggregate caliber distribution, total grape count distribution, mean caliber distribution, and the relationship between total grape count and mean caliber.

3.7. Stage 2: Histogram and Total-Count Prediction

The input representations defined above serve as the foundation for the second stage of the pipeline, where the objective shifts from geometric encoding to structured prediction. In particular, the RGB and RGBDT modalities provide complementary information that is exploited by the downstream models to infer both the global grape count and the caliber distribution. The following section describes how these representations are processed through a shared backbone and task-specific prediction modules. To ensure a consistent comparison across input modalities and prediction tasks, all models share a common feature extraction stage. This design isolates the effect of the input representation while maintaining a unified visual encoding, allowing differences in performance to be attributed to representation and modeling choices rather than architectural variability.

3.7.1. Shared Visual Backbone Architecture

All model families use the same visual backbone architecture, implemented through an ImageVarietyBackbone. The image branch is based on a pretrained MobileNetV3-Small.

The first convolution is adapted to accept either three or five channels, depending on the chosen modality. For the RGBDT configuration, the additional channels are initialized from the mean of the pretrained RGB filters.

In parallel, grape variety is encoded through a learned embedding. The visual descriptor and variety embedding are concatenated to produce a fused representation that is used by both the histogram and quantity models.

3.7.2. Residual Quantity Model

Total-count estimation is performed using a residual quantity model. Instead of predicting the total count directly from the image, the model receives the visual representation, the variety embedding, and the segmentation-derived base count. In addition to the raw count prior, the logarithm of the prior is also provided:

$$s = \text{seg_count_base}, \quad \log(1 + s).$$

The model predicts a residual correction,

$$r = f(\mathbf{x}, v, s),$$

and reconstructs the final total-count estimate as

$$\hat{N} = \max(s + r, 0).$$

3.7.3. Histogram Prediction Models

Six histogram-modeling families were evaluated, each trained under both RGB and RGBDT inputs: unimodal, bimodal, direct-bins, multitask-peak, CDF, and CDF-direct hybrid.

3.7.4. Output Reconstruction

Let $\hat{\mathbf{h}}$ denote the predicted normalized caliber histogram and $\hat{N}$ denote the predicted total grape count. The final predicted count per bin is computed as

$$\hat{\mathbf{y}} = \hat{N} \hat{\mathbf{h}}.$$

Once the model architectures are defined, the learning process is formulated through task-specific objectives that reflect both distributional accuracy and counting precision. The following section describes the loss functions used to supervise the histogram and quantity models, as well as their role in guiding the optimization process.

3.8. Training Objective

The quantity model is supervised using both the total-count target and the residual target:

$$r_{gt} = N_{gt} - s.$$

The histogram models are trained using a composite objective combining L1 loss on histogram probabilities, KL divergence, 1D Wasserstein distance, and L1 loss on the histogram-derived mean caliber. For the multitask-peak model, an additional cross-entropy term on the true peak bin is included.

3.8.1. Experimental Setup

All experiments were conducted on CUDA using an NVIDIA GeForce RTX 5080 GPU with 15.45 GB of VRAM. Training used a resolution of 512×512 , batch size 8, eight workers, and automatic mixed precision. In total, the benchmark included two residual

quantity models, one per input modality, and twelve histogram models corresponding to six modeling families evaluated under both RGB and RGBDT inputs. To ensure reproducibility, a fixed experimental protocol was adopted, including consistent data partitioning, identical backbone architectures, and controlled hyperparameters. The use of cached segmentation outputs ensures identical input representations across models, enabling fair comparison of modeling strategies. Furthermore, preliminary repeated runs indicated consistent trends across random initializations, suggesting stability of the reported results. While full statistical characterization was not included, these observations support the reliability of performance differences under the defined protocol.

The deployment experiments were conducted on a workstation equipped with an NVIDIA GPU using ONNX Runtime for inference. The exported models were evaluated without post-training optimizations, such as quantization or pruning, to preserve numerical fidelity with respect to the original PyTorch (version 2.1.0) implementation. The reported runtime reflects end-to-end inference, including preprocessing, segmentation, geometric encoding, and prediction stages. From a practical perspective, the compact model size and low latency suggest that the pipeline can operate in resource-constrained environments, such as edge or mobile platforms, provided that hardware acceleration is available. However, deployment remains subject to constraints related to memory, energy consumption, and robustness under variable acquisition conditions.

3.8.2. Evaluation Metrics

Histogram prediction quality was measured using histogram probability MAE, Wasserstein-1 distance, histogram count-space MAE, and mean caliber MAE derived from the histogram. Peak localization was measured using exact peak accuracy, peak ± 1 accuracy, peak ± 2 accuracy, and peak MAE in bins. Counting performance was evaluated through total-count MAE, total-count RMSE, residual MAE, and residual RMSE.

3.9. Ethics and Reproducibility

The proposed methodology was developed using image data collected under standard agricultural conditions, without involving human subjects or sensitive information, and therefore no ethical approval was required. The dataset consists exclusively of grape bunch images and associated annotations, without privacy concerns. To ensure reproducibility, all experiments followed a controlled protocol, including fixed data partitions, identical backbone architectures, and standardized training hyperparameters. Cached segmentation outputs and derived representations were reused across experiments, ensuring consistent inputs and enabling fair comparison across model variants. Furthermore, the implementation, including training scripts, configurations, and evaluation procedures, will be publicly released in a GitHub repository (https://github.com/ScaleAI-cl/paper_grapes_histogram_prediction, accessed on 1 June 2026) upon acceptance. This will facilitate independent verification and extension of the proposed framework. Additional documentation of experimental parameters and pretrained models will further support reproducibility and transparency.

4. Results

This section reports the performance of the two-stage pipeline. First, the results of the instance segmentation stage are presented, since this module provides both the geometry-aware inputs and the segmentation-derived counting prior used downstream. Second, the results of the histogram and total-count prediction stage are reported, including the comparison across RGB and RGBDT input modalities, the best-performing model, deployment-oriented analyses, and runtime measurements. All reported metrics

correspond to a single fixed experimental protocol; however, preliminary repeated runs indicated consistent trends across random initializations, suggesting stability under the defined protocol. However, statistical variability measures, such as standard deviation or confidence intervals, were not formally quantified and therefore remain part of future evaluation. In this context, the reported results should be interpreted as indicative of stable behavior rather than statistically validated performance differences. This reflects a cautious approach to ensure that conclusions remain aligned with the available evidence while acknowledging the need for more comprehensive statistical characterization in subsequent analyses.

4.1. Stage 1: Instance Segmentation Results

4.1.1. Training Dynamics and Error Patterns

Figure 6 shows the training dynamics of the instance segmentation model, including optimization losses and box- and mask-based metrics across epochs. Overall, the model exhibited stable convergence without major optimization instabilities.

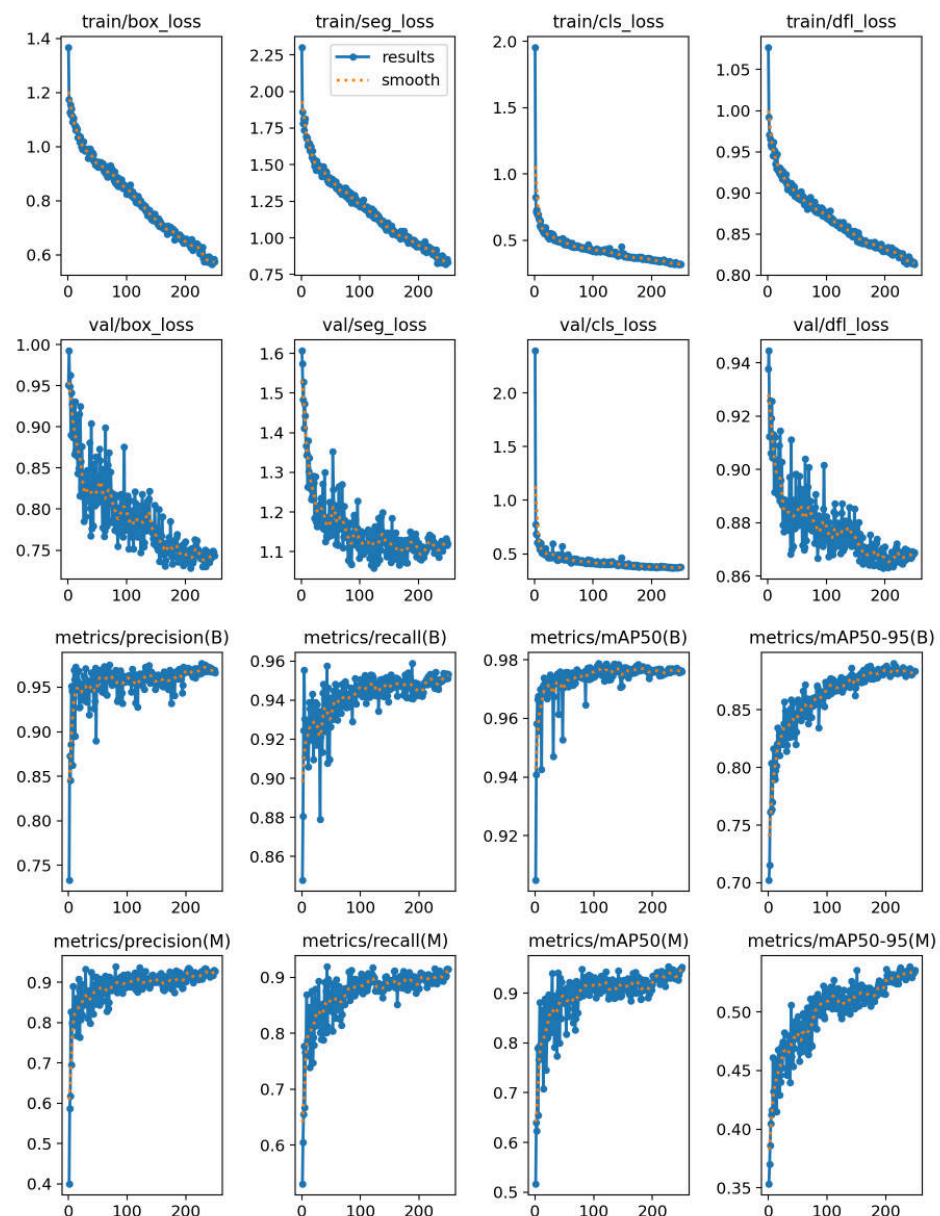


Figure 6. Instance segmentation training curves, including optimization losses and key box- and mask-based metrics across epochs.

Figure 7 presents the confusion matrix on the evaluation split. Most of the probability mass is concentrated along the diagonal, especially for pingpong and grape subclasses, while the dominant failure modes correspond to missed grapes predicted as background and background fragments incorrectly classified as grapes.

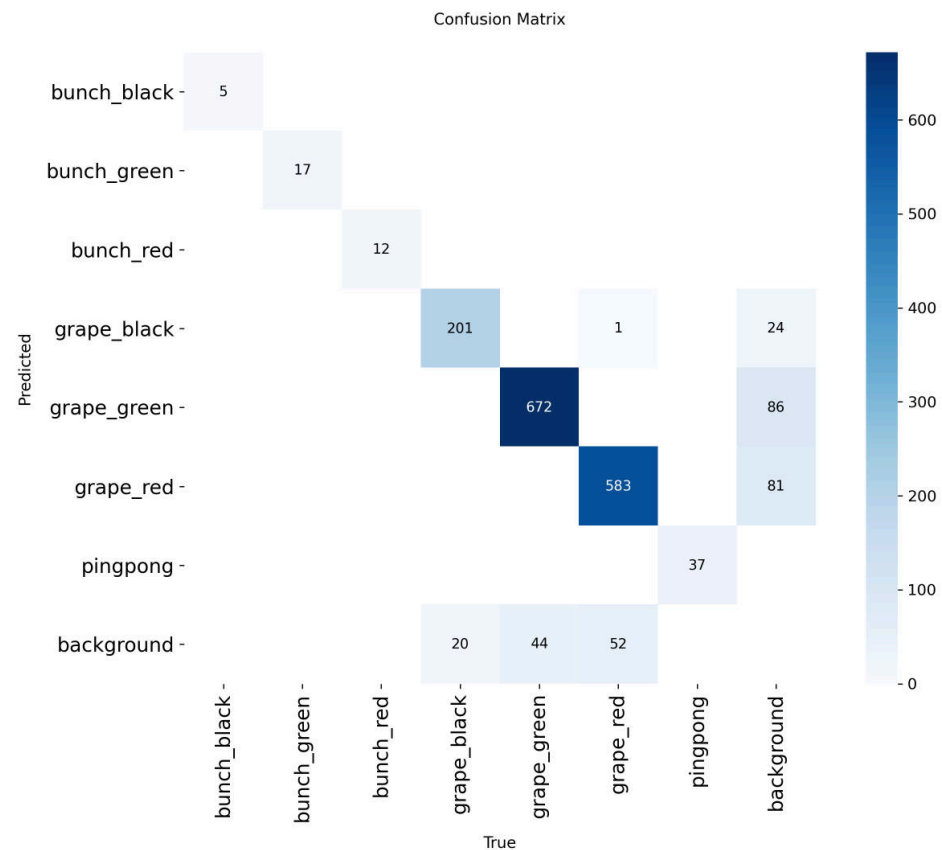


Figure 7. Confusion matrix for the instance segmentation classes.

4.1.2. Quantitative Segmentation Performance

Table 1 summarizes the performance of the YOLO-based segmentation model at an input resolution of 512×512 . Overall, the model achieves strong global performance, with box-level mAP@50 of 0.976 and mask-level mAP@50 of 0.935, indicating that detection and segmentation performance remains consistent under the evaluated conditions, including variability in illumination, viewpoint, and occlusion within the dataset. Notably, performance on the pingpong class is particularly high, reaching mAP@50 of 0.995 for both boxes and masks, with mask mAP@50–95 of 0.918. This is especially relevant, as accurate ping-pong segmentation directly supports the construction of the calibration-aware distance-transform channel. Furthermore, grape subclasses also exhibit strong mask-level performance, with mAP@50 values of 0.938, 0.959, and 0.946 for grape_black, grape_green, and grape_red, respectively. In this context, these results confirm that the segmentation stage is sufficiently reliable to support downstream geometric representation learning and residual count correction.

The high segmentation performance suggests that the extracted representations are of sufficient quality for downstream processing within the proposed pipeline; however, a direct causal relationship between segmentation metrics and histogram prediction performance is not explicitly established in this study.

Table 1. Instance segmentation performance of the YOLO-based model with input resolution 512×512 . Metrics are reported separately for bounding boxes and masks.

Class	Images	Inst.	P	R	mAP@50	mAP@50–95	P_m	R_m	mAP _m @50	mAP _m @50–95
all	35	1644	0.971	0.943	0.976	0.888	0.926	0.895	0.935	0.537
bunch_black	5	5	0.944	1.000	0.995	0.972	0.756	0.800	0.802	0.267
bunch_green	17	17	0.983	1.000	0.995	0.983	0.867	0.882	0.910	0.255
bunch_red	12	12	0.977	1.000	0.995	0.973	0.977	1.000	0.995	0.278
grape_black	5	221	0.968	0.825	0.947	0.745	0.963	0.816	0.938	0.655
grape_green	17	716	0.962	0.891	0.956	0.791	0.962	0.889	0.959	0.705
grape_red	12	636	0.969	0.883	0.948	0.770	0.964	0.878	0.946	0.679
pingpong	35	37	0.992	1.000	0.995	0.982	0.992	1.000	0.995	0.918

4.2. Stage 2: Histogram and Total-Count Prediction Results

4.2.1. Overall Comparison Across Histogram Families

Table 2 presents the ranking of the 12 histogram experiments obtained by combining six histogram-modeling families with two input modalities, RGB and RGBDT, evaluated on the fixed validation subset. A clear trend emerges: the top four positions are all occupied by RGBDT variants. This indicates that the geometry-aware distance-transform channels consistently improve downstream prediction relative to RGB-only input.

Table 2. Overall ranking of the histogram prediction experiments on the fixed validation subset. Lower “↓” is better for val_hist_w1 and val_count_total_mae, while higher “↑” is better for val_peak_exact.

Rank	Experiment	Input	val_hist_w1 ↓	val_peak_exact ↑	val_count_total_mae ↓
1	rgbdt_direct_bins	rgbdt	0.7563	0.3506	5.6659
2	rgbdt_cdf_model	rgbdt	0.7719	0.3665	5.6659
3	rgbdt_cdf_direct_hybrid	rgbdt	0.7752	0.3506	5.6659
4	rgbdt_multitask_peak	rgbdt	0.7762	0.3761	5.6659
5	rgb_cdf_direct_hybrid	rgb	0.7935	0.3581	5.9777
6	rgb_multitask_peak	rgb	0.7992	0.3676	5.9777
7	rgb_cdf_model	rgb	0.8031	0.3369	5.9777
8	rgb_direct_bins	rgb	0.8124	0.3570	5.9777
9	rgbdt_bimodal	rgbdt	0.8284	0.3347	5.6659
10	rgbdt_unimodal	rgbdt	0.8521	0.3242	5.6659
11	rgb_bimodal	rgb	0.8708	0.3189	5.9777
12	rgb_unimodal	rgb	0.9706	0.3008	5.9777

These results show that the best-performing configuration was the flexible direct-bin model combined with RGBDT input, rather than a more strongly constrained output parameterization. It is also worth noting that val_count_total_mae is shared across histogram families within each modality because a single residual quantity model is trained per input modality and then combined with all histogram variants of that modality.

Although the performance differences between RGB and RGBDT configurations are consistent, it is important to interpret these results beyond absolute metric values. The observed improvements suggest that the additional channels contribute information that is complementary rather than redundant, particularly in scenarios with occlusion or ambiguous boundaries. This indicates that the performance gain is not merely due to increased input dimensionality but to the explicit encoding of spatial structure derived from segmentation outputs. Such behavior reinforces the hypothesis that geometry-aware representations play a critical role in stabilizing distributional prediction under partial observability. The observed improvements associated with RGBDT inputs may arise from the inclusion of geometry-aware information; however, the current experimental design does not isolate

the effect of additional channels from potential dimensionality-related factors. A controlled ablation study would be required to disentangle these contributions.

4.2.2. Best-Performing Configuration

The best experiment in the current benchmark was `rgbdt_direct_bins`. Table 3 summarizes its validation performance. This model achieved the lowest Wasserstein distance among all evaluated configurations while maintaining low count-space histogram error and competitive peak-localization performance.

Table 3. Validation performance of the best histogram experiment, `rgbdt_direct_bins`, combined with the shared `qty_rgbdt` residual count model.

Metric	Value
<code>val_hist_prob_mae</code>	0.019158
<code>val_hist_w1</code>	0.756297
<code>val_mean_from_hist_mae</code>	0.682955
<code>val_hist_counts_mae</code>	1.136485
<code>val_peak_exact</code>	0.350636
<code>val_peak_pm1</code>	0.807203
<code>val_peak_pm2</code>	0.961864
<code>val_peak_mae_bins</code>	0.894068
<code>val_count_total_mae</code>	5.665889
<code>val_count_total_rmse</code>	8.691757
<code>underprediction_rate</code>	0.523305

Although `rgbdt_multitask_peak` achieved the highest exact peak accuracy, its Wasserstein distance was slightly worse than that of `rgbdt_direct_bins`. Therefore, the direct-bin RGBDT model provided the best balance between global histogram fidelity and count consistency. Preliminary repeated runs under the same experimental configuration indicated consistent trends across random initializations, with limited variability in key performance metrics. While full statistical characterization was not included in the current study, these observations suggest that the reported results are stable under the defined protocol.

4.2.3. Effect of Input Modality

A consistent advantage of RGBDT over RGB was observed across nearly all histogram families. For example, the direct-bin model shows consistently better performance under the evaluated protocol from `val_hist_w1` = 0.8124 in RGB to 0.7563 in RGBDT, while the bimodal model shows consistently better performance under the evaluated protocol from 0.8708 to 0.8284. This indicates that the segmentation-derived distance-transform channels encode geometric information that is useful beyond raw appearance alone, especially for separating touching berries and stabilizing histogram prediction.

The observed improvement associated with RGBDT input can be further interpreted in terms of representational enrichment rather than mere dimensional augmentation. The distance-transform channels provide explicit encoding of spatial proximity and object boundaries, which are otherwise implicitly inferred from RGB appearance. This additional structure is particularly beneficial in regions of high density and occlusion, where visual cues alone may be ambiguous. As a result, the model operates on a representation that more closely approximates the underlying geometric configuration of the scene, leading to consistently better performance under the evaluated protocol and improved stability in both histogram prediction and count estimation.

4.2.4. Effect of Histogram Parameterization

The comparison between histogram-modeling families showed that highly constrained output parameterizations were not optimal in this dataset. The unimodal and bimodal models ranked below the direct-bin, CDF, and hybrid variants, even under RGBDT input. This suggests that empirical grape caliber distributions are sufficiently flexible that direct or weakly constrained formulations provide a better fit than strongly imposed shape priors. At the same time, structured models were not uniformly poor. The CDF and CDF-direct hybrid variants ranked immediately below the best direct-bin model, suggesting that shape-aware formulations remain useful when they do not overconstrain the output distribution.

4.2.5. Residual Total-Count Estimation

The residual quantity model also benefited from RGBDT input. The best total-count performance reached $\text{val_count_total_mae} = 5.6659$ and $\text{val_count_total_rmse} = 8.6918$, compared with 5.9777 under RGB-only configurations. These results support the use of segmentation-derived counting priors followed by residual correction, rather than direct cardinality estimation from image features alone. The evaluation focuses on residual correction relative to a segmentation-derived counting prior. A direct comparison against a baseline model performing count estimation solely from image features was not included, and therefore conclusions should be interpreted within the scope of the proposed framework.

4.2.6. Qualitative Prediction Examples

Figure 8 presents six representative qualitative examples from the best-performing model, `rgbdt_direct_bins`. The selected cases cover three performance regimes: two best-case predictions, two intermediate cases, and two failure cases. This selection provides a more complete view of model behavior than best-case examples alone.

In the best cases, the model accurately captures the dominant caliber modes and preserves the overall distribution shape. In the intermediate cases, the predicted histogram generally follows the target trend, although some mass is redistributed across neighboring bins. In the worst cases, the main discrepancies are associated with peak displacement, local over- or underestimation, and reduced fidelity in the tails of the distribution.

4.2.7. Model Size and ONNX Fidelity

Despite the two-stage design, the deployed models remained compact. The `qty_rgb` and `qty_rgbdt` models contained approximately 1.11 million parameters, while the best histogram model, `rgbdt_direct_bins`, contained approximately 1.15 million parameters. The exported ONNX files remained below 0.5 MB, which is favorable for lightweight deployment.

ONNX Runtime validation also showed strong agreement with the original PyTorch models. For the RGBDT quantity model, the ONNX version produced a mean absolute difference of 0.023678 relative to PyTorch. For the best histogram model, the difference was negligible, with $\text{MAE} = 1.88 \times 10^{-5}$ and maximum absolute difference $= 2.227 \times 10^{-4}$. The small numerical differences observed between ONNX and PyTorch outputs do not produce noticeable deviations in downstream histogram or count metrics under the evaluated conditions, supporting the practical equivalence of both implementations for deployment purposes.

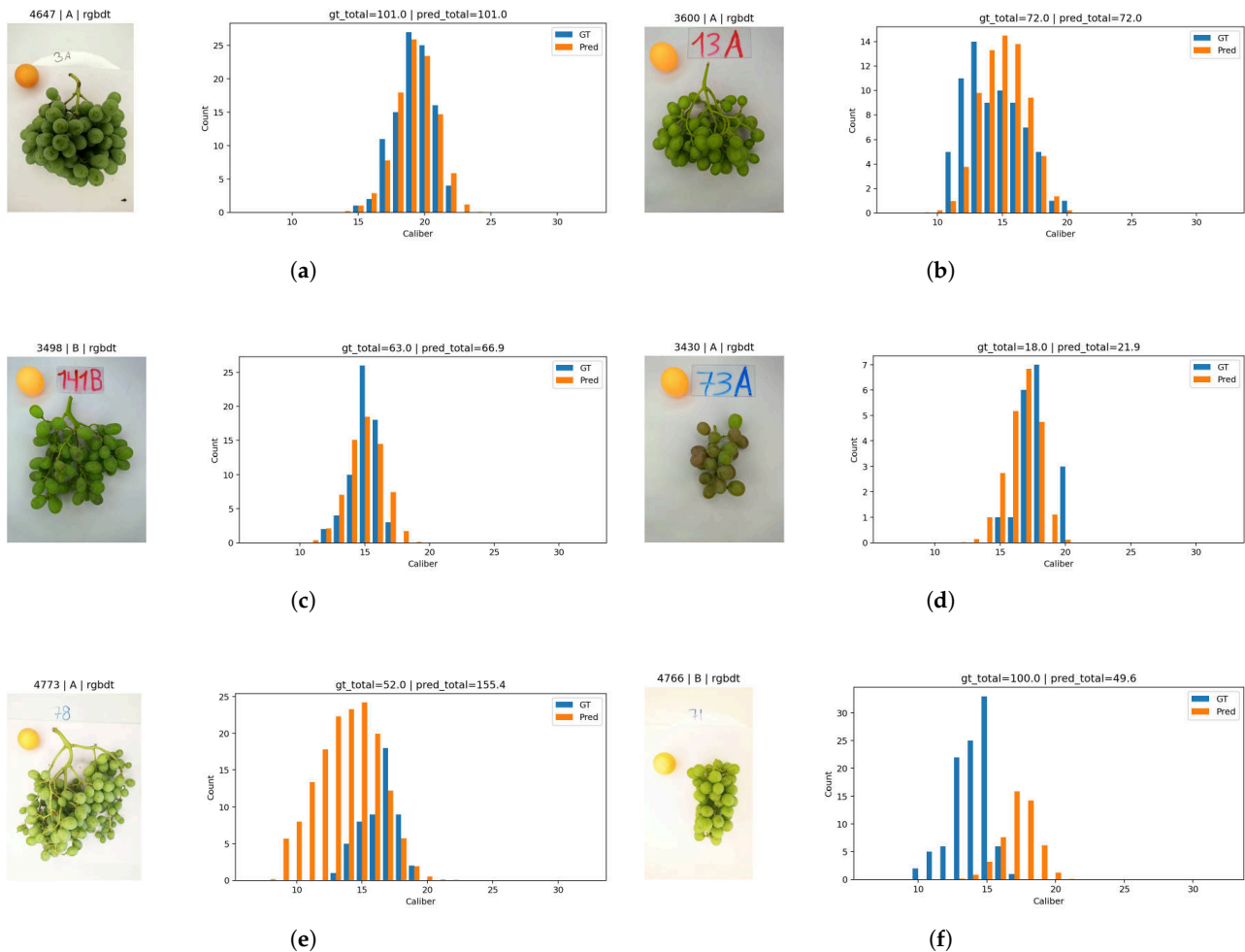


Figure 8. Qualitative prediction examples from the best-performing model (rgbdt_direct_bins). (a) Best case 1; (b) Best case 2; (c) Intermediate case 1; (d) Intermediate case 2; (e) Failure case 1; (f) Failure case 2.

4.2.8. End-to-End Runtime

The complete validation-time pipeline, including image loading, segmentation inference, channel construction, quantity inference, and histogram inference, achieved a mean runtime of 0.1240 s per image. Segmentation was the dominant cost, with a mean inference time of 0.0780 s per image, followed by channel construction at 0.0423 s. In contrast, the quantity and histogram models were lightweight, requiring on average 0.0022 s and 0.0015 s, respectively. Therefore, the learned predictors add little computational overhead relative to the segmentation stage.

5. Discussion

The results support four main conclusions. First, the instance segmentation stage is sufficiently reliable to support the downstream pipeline, both as a source of geometry-aware inputs and as the origin of the segmentation-derived counting prior. Second, geometry-aware channels derived from segmentation consistently improve histogram prediction and total-count estimation. Third, flexible histogram parameterizations outperform strongly constrained ones in this dataset. Fourth, residual count correction is an effective strategy for exploiting segmentation outputs without forcing the model to reconstruct cardinality from scratch.

5.1. Segmentation as an Enabling Stage

The proposed framework depends on segmentation not only for object localization but also for the construction of the grape and ping-pong distance-transform channels and for the computation of the base counting prior. In this context, the segmentation results are important because they validate the feasibility of the two-stage design. The high mask-level performance, especially on the pingpong class and grape subclasses, indicates that the downstream models receive sufficiently reliable geometric cues to support histogram and count prediction. At the same time, the segmentation results also clarify the main sources of downstream error. The confusion patterns suggest that the most relevant segmentation failures are missed grapes and background fragments confused as grapes [25]. These failure modes are not isolated: they can propagate to the RGBDT channels and to `seg_count_base`, thereby affecting both the histogram branch and the residual quantity model. Thus, segmentation should be interpreted as an enabling stage whose quality directly conditions the performance ceiling of the full pipeline.

5.2. Influence of Geometry-Aware Input Channels

The most consistent trend across the experiments was the superiority of RGBDT over RGB. All top four-ranked histogram models used RGBDT inputs, and each histogram family improved when the distance-transform channels were included. This indicates that the geometric structure encoded by the grape and ping-pong distance transforms provides information that is not recoverable from RGB appearance alone.

A likely explanation is that these channels make local object extent, separation between touching berries, and calibration-reference structure more explicit. This is particularly useful in dense bunches, where appearance alone may be insufficient to distinguish adjacent berries or to preserve shape-related cues relevant for caliber prediction. Importantly, these gains were obtained without introducing large downstream models, since the additional channels are lightweight once segmentation has been computed.

5.3. Why Direct-Bin Prediction Performed Best

A second important finding is that the best overall model was `rgbdt_direct_bins`, rather than a unimodal or bimodal alternative. This suggests that the empirical caliber histograms in the dataset are not adequately described by overly restrictive shape assumptions. Real bunches may exhibit asymmetric, broad, skewed, or mildly multimodal distributions that benefit from direct prediction of bin probabilities.

The competitive performance of the CDF and CDF-direct hybrid variants further supports this interpretation. Some amount of output structuring appears helpful, especially because caliber bins are ordered, but excessive parametric restriction reduces fidelity. In this sense, the direct-bin RGBDT model offers the best compromise between flexibility and stability, preserving the ability to represent irregular histogram shapes while still learning from geometry-aware input cues.

5.4. Value of Residual Count Learning

The residual quantity model also proved to be a strong design choice. Rather than replacing segmentation-based counting, the model uses the number of detected grapes as a prior and learns to correct its systematic bias. This is advantageous because segmentation already captures a large amount of counting information but tends to undercount when berries are occluded, merged, truncated, or fragmented.

By learning only the missing correction, the quantity branch solves a simpler problem and reaches a lower total-count error than a direct cardinality estimator would likely achieve under the same conditions. This strategy also integrates naturally with the his-

togram branch, since the final count-per-bin prediction depends directly on a reliable estimate of total bunch cardinality. This formulation aligns with established residual-learning approaches in regression and counting tasks, where coarse predictions are refined through learned corrections to improve robustness and stability under noisy or incomplete observations [26].

5.5. Insights from Qualitative Examples

The qualitative examples shown in Figure 8 complement the aggregate metrics by illustrating model behavior across varying difficulty levels. In the best cases, the model captures dominant caliber modes, preserves distribution spread, and closely matches total count, indicating that reliable segmentation-derived inputs support accurate reconstruction of both shape and scale. In intermediate cases, the global trend is maintained, although mass is redistributed across neighboring bins, suggesting that residual errors arise from local ambiguity rather than misinterpretation. From an application perspective, such deviations are less critical than large count errors or distribution shifts. In contrast, failure cases exhibit peak displacement, over-smoothing, and count misestimation. In some instances, qualitative inspection suggests mismatches between visible structure and annotations, likely reflecting labeling challenges under occlusion rather than systematic errors. Overall, performance degrades under occlusion, high density, and segmentation uncertainty.

5.6. Analysis of Hypotheses and Research Questions

The experimental results provide consistent evidence to analyze the proposed hypotheses and research questions from both performance and modeling perspectives. With respect to RQ1, the two-stage monocular pipeline enables stable estimation of total grape count and caliber histograms under realistic conditions, suggesting that separating perception and inference helps address partial observability. Regarding RQ2, the comparison between RGB and RGBDT indicates that geometry-aware channels improve both histogram fidelity and count estimation, likely by encoding spatial cues not captured by RGB alone. For RQ3, residual correction enhances robustness by compensating systematic biases in segmentation-derived priors. From a modeling perspective, flexible approaches better capture irregular distributions. Furthermore, residual learning aligns with established correction-based frameworks in regression and counting tasks [27] that refine coarse predictions to improve stability under uncertainty [28]. Overall, the results highlight a robust yet constrained framework, with remaining challenges linked to occlusion and segmentation uncertainty.

5.7. Practical Implications for Deployment

From a deployment perspective, the reported runtime results are encouraging. The learned histogram and quantity models are lightweight, compact, and faithfully exportable to ONNX. The dominant computational cost lies in the segmentation stage, not in the downstream predictors. Therefore, future optimization efforts for real-time or embedded deployment should focus primarily on accelerating the segmentation network or reducing the cost of channel construction. This is a practically favorable outcome. Since the histogram and quantity models contribute only a small fraction of the total runtime, the method remains deployable even when multiple histogram parameterizations are explored during development. The compact ONNX footprint also supports use in phenotyping and quality-control settings where lightweight inference is desirable [29]. From an application perspective, the proposed framework can be integrated into existing viticultural workflows through mobile or edge-based acquisition systems. For instance, the lightweight ONNX models and low runtime suggest feasibility for deployment on handheld devices or field robots, enabling near real-time estimation during harvesting or inspection [30]. Such integration would require minimal user interaction, relying on

single-image capture and automated inference, thereby aligning with practical constraints in agricultural environments.

5.8. Limitations

Despite favorable results, several limitations remain. First, the effectiveness of the residual count prior depends on segmentation quality, where occlusion, mask fragmentation, or domain shift may degrade both auxiliary representations and base counts. Second, histogram targets rely on annotation quality, and limited label noise may influence training and evaluation. Third, the evaluation is conducted on a single dataset under bounded conditions and therefore does not explicitly assess cross-domain generalization across vineyards, seasons, or acquisition setups. Furthermore, although RGBDT consistently outperforms RGB, its benefits depend on the robustness of the segmentation stage, and errors may propagate through geometric encoding. From a methodological perspective, the reuse of a shared quantity model ensures that histogram comparisons isolate parameterization effects while maintaining consistent count estimation. Overall, these findings highlight the need for improved robustness, interpretability, and cross-domain validation in future work.

5.9. Future Work

Future work should explore several directions. First, improving segmentation robustness may enhance both total-count estimation and histogram accuracy. Second, uncertainty-aware histogram prediction could provide confidence estimates in ambiguous or low-support regions. Third, multi-view fusion may improve robustness by mitigating self-occlusion and partial visibility in dense bunches. Furthermore, evaluating the framework across diverse acquisition conditions and domains is necessary to better characterize generalization. Overall, the results indicate that combining segmentation-derived geometric cues, residual quantity correction, and flexible histogram modeling provides an effective and deployable strategy for grape caliber analysis. From a practical perspective, future work will also explore quantization and hardware-specific optimizations to further reduce latency and energy consumption for embedded deployment.

6. Conclusions

This study introduced a two-stage monocular pipeline for grape caliber analysis from single images. The proposed approach combines instance segmentation, segmentation-derived geometric channels, residual quantity correction, and histogram prediction within a lightweight and deployable framework.

The results show that the segmentation stage provides a sufficiently reliable basis for downstream prediction, both for geometry-aware feature construction and for the initialization of the counting prior. They also show that RGBDT inputs consistently improve performance over RGB alone and that the best-performing model is `rgbdt_direct_bins`, indicating that flexible direct-bin histogram prediction is better suited to the observed grape caliber distributions than more constrained unimodal or bimodal formulations. In addition, residual learning over a segmentation-derived count prior proved effective for total-count estimation, improving the final reconstruction of count-level histograms. The effectiveness of residual correction is demonstrated relative to a segmentation-derived counting prior; however, a direct comparison against a baseline model performing count regression solely from image features was not included in the current study. Therefore, the reported improvements should be interpreted as evidence of consistency within the proposed framework rather than as a definitive comparison against all possible counting strategies.

From a deployment perspective, the exported ONNX models preserved numerical fidelity, and the complete pipeline achieved lightweight runtime, with segmentation remaining the dominant computational cost. Although the method still depends on segmentation quality and some difficult samples may include substantial prediction errors or possible annotation discrepancies, the overall results support the proposed design as a practical strategy for monocular grape caliber estimation. Some discrepancies observed in qualitative examples may reflect inherent challenges in defining precise ground-truth distributions under dense and partially occluded conditions, rather than systematic annotation errors.

Future work should address segmentation robustness, uncertainty-aware prediction, multi-view extensions, and broader cross-domain evaluation. Importantly, these directions are presented as prospective developments rather than conclusions directly supported by the current evidence, reflecting a deliberate effort to guide further investigation. Beyond this scope, the proposed framework illustrates a broader paradigm for structured prediction from monocular imagery, combining perception, geometry-aware representations, and distribution-level inference within a lightweight architecture. This design enables both local object understanding and global aggregate estimation under partial observability. Furthermore, the integration of efficient perception models with structured prediction highlights a scalable approach for deployable agricultural systems, where robustness, computational efficiency, and adaptability are essential. Overall, this perspective suggests a transferable design principle for similar tasks involving dense object arrangements and constrained operational environments.

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Abbreviations

The following abbreviations are used in this manuscript:

RGB	Red, Green, Blue image representation
RGBDT	RGB with Distance Transform channels
DT	Distance Transform
ONNX	Open Neural Network Exchange

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